Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/12/2014
PI Name:	Hollberg, Leo W Ph.D.		
Project Title:	Advancing Time Transfer and Optical Atomic Clocks for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lhollberg@earthlink.net	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-582-4568
Organization Name:	Stanford University		
PI Address 1:	Physics Department		
PI Address 2:	452 Lomita Mall, HEPL		
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	05/30/2012	**End Date:**	05/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1458931		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our research project focuses on one of the most critical and significant scientific and technical challenges to using advanced cold-atom atomic clocks in space. The problems that we address are the serious performance limitations to the existing Time-Transfer links between ground and space. This research will advance both the basic science and the technology of Time-Transfer methods that could both support the current ESA-ACES mission plan and subsequently enhance performance in order to meet the demands of future optical atomic clocks (e.g. ESAs-SOC) for space. The improved performance is enabled by advanced laser methods for absolute Time and Frequency transfer that are based on stabilized femtosecond mode-locked lasers. We have developed a novel integrated system concept that will optimize the Time-Transfer performance in real-world systems by combining an ultrahigh performance Laser Time-Transfer (Laser-TT) link with an advanced microwave link and GPS signals. Analysis of the fundamental capabilities and limitations will guide the final system design, and a prototype Laser-TT link will be built in the laboratory and tested to		

	critically evaluate the system performance. The demonstrations and testing of the Time-Transfer link will include two-way laser transmission via both free-space and optical fibers, and two-way microwave transmission via coaxial cable, which will give over-constrained redundant signals allowing us to identify systematic errors and uncertainties. Our Laser-TT link will improve the absolute timing precision by a factor of 100 times compared to the most advanced microwave links. Nonetheless, the highest performance, practical and reliable systems operating in real-world conditions (with weather, turbulence, etc.) will require both microwave and optical channels. Critical analysis and testing of the most promising concepts will allow us to optimally integrate an advanced microwave link (as per ACES) with our higher performance Laser-TT link.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced precision for "Time" (epoch), time interval, and frequency transfer to space could be enabling for several science and technology missions and applications, including: ISS, ACES, tests of relativity, geodesy, gravity. Optical timing can improve the performance and precision of Global Satellite Navigation Systems (GNSS) such as GPS. It could also provide much improved precision and accuracy of precise orbit determination and through better timing and orbit could enhance capabilities and accuracy in geodesy. Potentially leverage capabilities with existing International Laser Ranging Systems to mutual benefit. Definitely strong synergism with satellite laser communication systems now under development.
Task Progress:	This project has focused on developing laser methods to transfer precise Time and frequency signals through free-space optical links. Precision timing is critical for several space science missions, particularly those looking to test Einstein's General Relativity (e.g.: measurements of the gravitational Red Shift, tests of local Lorentz Invariance, tests of Local Position Invariance, missions to detect Gravitational waves), that could provide new capabilities and enhance performance of GNSS navigation systems and applications in geodesy and Earth sciences. Part of this effort—the development of optical timing systems that can link high performance clocks on the ground to clocks in space—is in support of the upcoming ACES mission of ESA. ACES will have a corner-cube retro-reflector array and an optical event timer on board that can time received optical pulses relative to the local ACES clocks; the retro-reflected pulses detected on the ground will give the round trip time delay. Our baseline approach for optical timing is to use a fs optical frequency comb that is referenced to atomic frequency standards and then encode and transfer the precisely timed optical pulse over a two-way optical link. In this phase of the effort we have focused on frequency stabilization of the comb, evaluating different optical link architectures, and initial tests of novel pointing and tracking methods.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Abstracts for Journals and Proceedings	Hollberg L, Berceau P. "Laser Link for Space-Time Referencing." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 49. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Papers from Meeting Proceedings	Berceau P, Hollberg L. "Laser time-transfer and Space-Time Reference in orbit." Presented at the Sixth Meeting on CPT and Lorentz Symmetry, Bloomington, Indiana, June 17-21, 2013. Proceedings of the Sixth Meeting on CPT and Lorentz Symmetry (CPT'13). World Scientific, 2014. See also http://arxiv.org/abs/1402.1207 , Mar-2014